

# A cup of MOCCa

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## I. INTRODUCTION: TANTALUS AND HEPHAESTOS

Tantalus in its current incarnation is extremely versatile: the code (i) offers a great many options for the user to impose (or not) different selfconsistent symmetries and (ii) a (in principle) very large variety of different forms of the Skyrme EDF. Offering a combination of all of these possibilities in one executable (or equivalently, one set of source code files) is **very** complicated, as I found out with MOCCav1. The other extreme, writing many separate codes becomes a maintenance nightmare and bugfixing party, as I found out during my early days in the PNTPM at ULB. This is why the architecture of Tantalus, MOCCav2, is very different: as much as possible of what I call "complicated choices" (comprising, but not limited too, the specifics of the form of the EDF and symmetry options) are decided **compile-time**. There exists a meta-code in my private repository, called Hephaestos, that writes appropriate Tantalus code based on a particular instance of "complicated choices". In pseudo-code:

```
python Hephaestos < Hephaestos.in > [Tantalus code]
```

Different inputs to Hephaestos produce different source code for Tantalus, which can then be compiled into a separate executable. This manual does not document any aspect of Hephaestos, instead focussing on the commonalities and differences of the multiple resulting sets of Tantalus source code.

## II. COMPILATION AND ALL DIFFERENT SOURCE CODE SETS

There are currently four different code-sets in the repository. These are:

- **src/** : Default option, maximally symmetric executable, named `Tantalus.NLO.func.exe` by default.
- **src\_T/** : Time-reversal breaking, but spatially maximally symmetric. `Tantalus.NLO.func.T.exe` by default.
- **src\_P/** : Parity breaking, time-reversal symmetric. `Tantalus.NLO.func.P.exe` by default.
- **src\_TP/** : Parity and time-reversal breaking, named `Tantalus.NLO.func.TP.exe` by default.

More information is summarized in Table I. The makefile provided with the repository allows for easy compilation of all of these. A simple `make` command will compile **all** of these executables, but individual ones can be compiled with the commands listed in the first column of Table. I. The compiled executables can all be found in the `exec/` folder. By default, the code is compiled with the `gfortran` compiler. While this is very portable and compiles comparatively quickly, this is a **suboptimal** choice for production use: at least on the machines available in Brussels, compiling with **ifort** results in executables that are 20% faster or more. One can specify the compiler to the makefile, by typing

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```
make CXX=ifort
```

If you feel confident, you can add extra flags to the compilation process by using the `CXXFLAGS` flag, but I suggest reading the Makefile first in that case.

Since the different executables and compileoptions can quickly give rise to confusion. For this reason, the Makefile will include some information on the actual compilation options in the header of the code, such as for example:

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre> MOCCa v2.0 =  #####  ##  #  # #####  ##  #  #  #  ##### #  #  #  ##  #  #  #  #  #  #  #  #  # #  #  #  #  #  #  #  #  #  #  #  #  ##### #  #####  #  #  #  #####  #  #  #  # #  #  #  #  ##  #  #  #  #  #  #  #  # #  #  #  #  #  #  #  #  #  #####  #####  #####  Copyright P.-H. Heenen, M. Bender &amp; W. Ryssens  ----- Runtype = Single-mode ----- Version Information -----  "commit bb9a0523b8b85c571e30ea524aa3bac47901db42" "Author: Wouter@Orome &lt;wouter.ryssens@gmail.com&gt;" "Date: Tue Jan 11 20:47:32 2022 +0100"  ----- Symmetry Information -----  S.p. generators      = Rz,T,STy Axis reduction  X Y Z = 1 1 0 SYM_CODE           = 0 0 001 000 10 000 010 110 TRANS_CODE         = 0 1 001 000 10 000 010 111  ----- Environment Information ----- ----- Compilation Information -----  Compiled with: "GNU Fortran (Ubuntu 9.3.0-17ubuntu1~20.04) 9.3.0" Compilation flags reported: "-O3 -Jmod -Wall -Wno-uninitialized "</pre> | <pre> =&gt; Deprecated information  =&gt; Precise git commit reference  &gt; Symmetries imposed on the nucleus &gt; can be combination of Rz, P, T, STy &gt; Axis reduction: which axes are fully (0) &gt; represented or not (1). &gt; Codes referring to input and output of wfs &gt; Not used yet  &gt; output of "\$CXX --version"  &gt; CXXFLAGS; compilation options</pre> |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

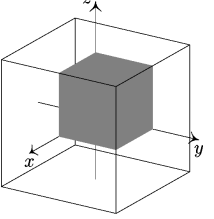
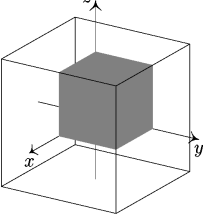
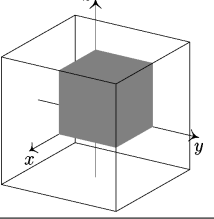
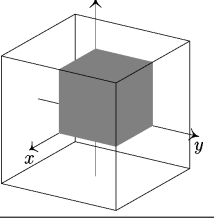
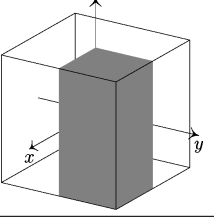
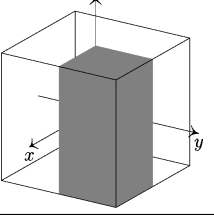
| Tantalus.XXX.exe                      | source folder | mesh & spwfs                                  | Typical use                                                                            | Spatial representation                                                                |
|---------------------------------------|---------------|-----------------------------------------------|----------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| NLO.func<br>make single               | src           | $(N_x/2, N_y/2, N_z/2, dx)$<br>(nwn/2, nwp/2) | Starting point<br>Multipole moments: oriented around Z-axis                            |    |
| NLO.func.crankX<br>make single_crankX | src_crankX    | $(N_x/2, N_y/2, N_z/2, dx)$<br>(nwn/2, nwp/2) | Starting point for cranking calculations<br>Multipole moments: oriented around X-axis  |    |
| NLO.func.T<br>make single_T_crankX    | src_T         | $(N_x/2, N_y/2, N_z/2, dx)$<br>(nwn, nwp)     | Nuclei with odd-N/Z<br>Multipole moments: oriented around Z-axis                       |    |
| NLO.func.T.crankX<br>make single_T    | src_T_crankX  | $(N_x/2, N_y/2, N_z/2, dx)$<br>(nwn, nwp)     | Rotational bands<br>Multipole moments: oriented around X-axis                          |   |
| NLO.func.P<br>make single_P           | src_P         | $(N_x/2, N_y/2, N_z, dx)$<br>(nwn/2, nwp/2)   | Octupole deformation and fission barriers<br>Multipole moments: oriented around Z-axis |  |
| NLO.func.TP<br>make single_TP         | src_TP        | $(N_x/2, N_y/2, N_z, dx)$<br>(nwn, nwp)       | Octupole for odd-Z/N<br>Multipole moments: oriented around Z-axis                      |  |

TABLE I. Summary of different source code sets currently available: default executable names, source code folders and (mesh,single-particle) number parameters are given with respect to a calculation with no symmetries imposed.

### III. RUNNING TANTALUS

Any Tantalus executable can be used as

```
./Tantalus.[.....].exe < tant.in
```

This simple line masks however that there are multiple inputs and outputs.

#### Input

- 
- (1) tant.in : REQUIRED  
An input file detailing the specifics of the calculation, such as nucleon number, name of the parameterization, mesh discretisation, ... This is fed to the executable through STDIN.
  - (2) \${PARAM}.param file : REQUIRED  
An input file containing the requested the parameter values of the EDF. Must be present in the current directory where Tantalus is executed. Can be either a file named as
    - \${param}.param: named after the parameterization, ex. SLy4.param, containing a SINGLE parameter set.
    - forces.param : a file containing multiple parameter sets.
 Note, the code will look first for files names \${param}.param and will only look for forces.param if the more specific file cannot be found.
  - (3) \${INPUTFILENAME}.wf : OPTIONAL  
A wavefunction (.wf) file created by some previous Tantalus run to warmstart the calculation. Must be present in the current directory where Tantalus is executed.

#### Output

- 
- (1) STDOUT : ALWAYS PRODUCED  
A wealth of information on all kinds of quantities, from the first iteration to the last. If it is calculated by the code, then it is printed here. All other outputs (with the exception of the single-particle wavefunctions) are reformats for convenience of information that can in principle be gotten from here. Can be captured through piping or the 'tee' command on Unix.
  - (2) \${OUTPUTFILENAME}.wf : ALWAYS PRODUCED  
A .wf file containing the final state of the code on exit. Useful for warmstarting the next calculations.
  - (3) \${BXLFIT}zXXXnYYY.out: NOT PRODUCED BY DEFAULT  
A one-line text file containing observables from the final iteration of the code. Useful for automatic combining many calculations into, for instance, a parameter fit. Name (and presence) determined by the 'BXLFIT' parameter discussed below.
  - (4) \${DENFILE} : NOT PRODUCED BY DEFAULT  
File containing the neutron, proton and charge density at all the mesh points.
  - (5) \${POTFILE}: NOT PRODUCED BY DEFAULT  
File containing the neutron and proton potential associated with the density, as well as Coulomb potential for both nucleon species at all the mesh points.
  - (6) \${SPHFFILE} : NOT PRODUCED BY DEFAULT  
File containing the properties of the single-particle wavefunctions in the Hartree-Fock basis.
  - (7) \${SPCANFILE}: NOT PRODUCED BY DEFAULT  
File containing the properties of the single-particle wavefunctions in the canonical basis.

## IV. INPUT

|                    |                                                                           |
|--------------------|---------------------------------------------------------------------------|
| &nucleus           | / ! Information on the nucleus and convergence                            |
| &mesh              | / ! Mesh parameters                                                       |
| &func              | / ! Name of the parameterization                                          |
| &pairing           | / ! Options linked to the treatment of pairing (and blocking!)            |
| &indices           | / ! Options and indices for blocking quasiparticles                       |
|                    | ! NOTE: Optional                                                          |
| &evolution         | / ! Options related to the evolution of the single-particle wavefunctions |
| &scfiteration      | / ! Options for the SCF procedure: mixing of densities, potentials etc... |
| &wfs               | / ! Number of wavefunctions and options for transformation                |
| &IO                | / ! Input and output options                                              |
|                    | ! (i) multiple kinds of filenames                                         |
|                    | ! (ii) symmetry transformation options                                    |
| &Inertia           | / ! Determines for which l,m the collective inertias get calculated       |
| &MomentParam       | / ! General treatment of multipole constraints                            |
| &MomentConstraint/ | ! Specify a single ( = one,1 ) multipole constraint.                      |
|                    | ! NOTE: optional, can be repeated                                         |
| &Cranking          | / ! Information on cranking and determining angular momentum              |

FIG. 1. List of namelist inputs, in the correct order.

## A. The STDIN input: tant.in

Tantalus reads input from STDIN, using the FORTRAN namelist machinery. The available namelists are indicated in Fig. 1; the ordering of the namelist is important and all of them are required. The only exceptions to this are (i) the &indices/ namelist, (ii) the &momentconstraint/ namelist and (iii) the &inertia/ namelist. All are only required if the user activates other parameters in the &pairing/, &MomentParam/ and &IO/ namelists. In addition, &momentconstraint/ can be repeated. Below, I document the input parameters in all of these namelists; please do not assume that all of the below is complete.

I have tried to indicate parameters that are either experimental and/or are unlikely to be useful for most people's use by "A" (for advanced) in the first column. I do not guarantee that these work correctly (although almost all do).

## &amp;nucleus

```

-----
neutrons    = Number of neutrons. Can be non-integer.
protons     = Number of protons. Can be non-integer.
energy_prec = convergence criterion on the energy.
moment_prec = convergence criterion on the multipole moments.
disp_prec   = convergence criterion on the weighted dispersion of the spwfs.
fermi_prec  = convergence criterion on the Fermi energy.
A pairing_prec = precision required of the pairing solver.
               [Recommendation: DO NOT TOUCH!]
A inversetemp = inverse temperature, in units of MeV.
A fixfermi   = Logical, if .true. keep the Fermi energies constant.
A mun, mup   = Fermi energy for neutrons/protons, units of MeV.
-----

```

## &amp;mesh

```

-----
nx/y/z      = number of points in x/y/z direction.
dx          = mesh discretisation (in fm).
-----

```

## &amp;func

```

-----
name_param  = name of the parametrization, needs to match a name on forces.param.
-----

```

`&pairing`


---

```

Type           = 'HF', 'BCS' or 'HFB'.
Guesssgaps     = Logical. If .true. initialize a fixed starting guess for Delta's.
BlockType      = 0: No blocking
                1: True blocking based on indices
                2: True blocking based on lowest energy,
                3: EFA based on indices
                4: EFA based on lowest energy
BlockNumber    = Number of quasiparticles to excite.
                If Blocknumber > 0, read &Indices/ namelist after &pairing.
pairingscheme  = 0 : direct solver   = fast but often unstable for odd-Z/N
                1 : gradient solver = slow but almost guaranteed to converge
Bogofromfile   = Logical. If .true. (= default value), use the Bogoliubov
                transformation read from file to continue the calculation.
A  maxhfbiter  = Maximum number of iterations for the Fermisolver at each
                mean-field iteration. (Default = 200)
A  FermiSolver  = Numerical solver to use to determine the Fermi energies
                "Brent" => New and stable (default).
                "Secant"=> Old and not so stable.
A  gradient_precon = Logical. If .true., use the gradient preconditioning scheme of
                the Madrid group. Only active if pairingscheme = 1.
A  Particles_in_gas = Integer tracking the way the contribution from a free gas
                is counted for the readjustment of the Fermi energy. Note
                that this is only active at finite temperature and when
                using BCS!
                0) No correction: the number of particles is the ordinary counting, Tr(rho).
                1) Subtraction : the nucleus is modelled as being immersed in a gas of
                    free particles. The number of particles is Tr(rho) MINUS
                    the number of particles in a free gas at the same chemical
                    potential.
                2) Nogas      : the code is only allowed to have non-zero occupation
                    numbers for single-particle states with negative
                    single-particle energy.
A  Lambda_2    = Multiplier values for constraint on the dispersion of the
                particle number. Default: (0.0,0.0).

```

---

`&Indices` (NOTE: this namelist is only read if BlockNumber > 0)

---

```

BlockIndices = Blocknumber integers. These are the indices of the states in the
                Hartree-Fock basis, for which the code will attempt to construct
                quasiparticle excitations with large overlap with them.
BlockLowest  = Set of 'a2' formatted strings, of length BlockNumber.
                'n+', 'n-' : lowest quasi-neutron of positive/negative parity
                'p+', 'p-' : lowest quasi-proton of positive/negative parity
                'n0', 'p0' : lowest quasi-neutron/proton of either parity.
Note that
(a) all of these can be combined. The input
    'n+', 'n+', 'p0', 'p0'
    will result in the code trying to construct the lowest state
    with two-quasi-neutrons of positive parity and two
    quasi-protons of whatever parity.
(b) If time-reversal is broken, this option will only consider
    blocking options with positive signature. (Which makes no
    difference if time-reversal is conserved)

```

---

## &amp;evolution

---

maxiter = maximum number of mean-field iterations.  
 printiter = Large printouts after every multiple of this value.  
 Attention: collective corrections are only recalculated at every total printout, as determined by this parameter.  
 gradient\_safety = Safety parameter for the estimation of the parameters of the gradient scheme in units of MeV. Only active when pairingscheme=1. Default: 0.1. Extra-safe value: 2.  
 A Strategy = Evolution strategy for the single-particle wavefunctions.  
 "HEAVYBALL" : heavy-ball scheme, default.  
 "IMTIME" : imaginary time-evolution.  
 A dt, momentum = Parameters governing the heavy-ball update. Note: read values will be overwritten if EstimateParams=.true.  
 A Estimateparams = If .true. (= default), use heuristics to guess optimal evolution parameters dt and momentum at every iteration.  
 A gradient\_stepsize, gradient\_mu = Parameters governing the heavy-ball update. Note: read values will be overwritten if EstimateGradParams=.true. Only active if pairingscheme=1.  
 A EstimateGradparams = If .true. (= default), use heuristics to guess optimal evolution parameters for the gradient solver in the pairing space. Only active if pairingscheme=1.

---

## &amp;scfiteration/

---

A scfscheme = Integer determining the type of SCF evolution  
 0: preconditioning of F\_I\_I and F\_I\_S potentials  
 1: linear mixing of D\_I\_I and D\_I\_S densities  
 A denmix = Parameter governing the density mixing (Default: 0.75)  
 $D_{I_I\_new} = denmix * D_{I_I\_old} + (1 - denmix) * D_{I_I\_new}$   
 A preconfactor = Parameter governing the potential preconditioning  
 $F_{I_I\_new} = F_{I_I\_old} + P^{-1} (F_{I_I\_new} - F_{I_I\_old})$   
 $P^{-1} = 1 + preconfactor * Delta$

---

## &amp;wfs

---

nwn/nwp = number of neutron/proton wavefunctions.  
 Attention: these need to match the input .wf file, if present, modulo the symmetry transformation rules discussed below.  
 A osc\_freq = Harmonic oscillator frequencies in all Cartesian directions used for the generation of Nilsson orbitals if starting from scratch. (omega\_x, omega\_y, omega\_z) with typical size ~0.2 MeV.  
 Strictly spherical : omega\_x = omega\_y = omega\_z  
 Prolate, symmetry axis Z : omega\_x = omega\_y > omega\_z  
 Oblate, symmetry axis Y : omega\_x = omega\_z < omega\_y  
 Default: (/ 0.2125, 0.2125, 0.175 /).

---

&IO

---

Inputfilename = name of the .wf file Tantalus will read. If this is 'INIT',  
Tantalus will not read any file and initialize from scratch.

Outputfilename = name of the .wf file Tantalus will write.

BXLFIT = string. Extra output will be written to the following file  
'\${BXLFIT}zXXXnYYY.out'  
where XXX = the number of protons (forced to be integer)  
YYY = the number of neutrons (forced to be integer)  
\${BXLFIT}= the value of the string.  
The exact content of this file is explained below.

denfile = filenames for the write-out of extra information. In order:  
potfile scalar densities, mean-field potential, single-particle info  
sphffile in the HF basis and single-particle info in the canonical basis.  
spanfile Finally, the inertfile outputs the (asked-for) collective  
inertfile inertia parameters.

N\_inertia = number of collective degrees of freedom to calculate the  
inertias for. If non-zero, the code will read the  
&inertia/ namelist next.

Extraspwfs = 8 integers. If non-zero, the code will add this many (randomly  
generated) spwfs in the corresponding isospin-parity-signature  
blocks. Note that the nwn/nwp numbers need to be correctly  
adapted, meaning nwn = nwn\_file + number of new neutron spwfs.

AllowTransform= Logical. If .true., it allows the code to transform the input,  
meaning that you can either (a) add/remove mesh points  
(b) add extra spwfs (never remove!)  
(c) break a symmetry

A checkpointiter= Write a checkpoint (wavefunction file) to disk every time this  
many iterations have passed.

A tofile = filename for the writeout of time-odd densities D\_I\_S and  
C\_I\_N to a file with this name.

A blockfile = filename to write the densities  $|\psi|^2$  to of the blocked  
single-particle wavefunctions (in the canonical basis)

---

&Inertia \* Optional, only read if N\_inertia is > 0

---

Inertia\_l = lists of values l/m for collective degrees of freedom Q<sub>lm</sub> for  
Inertia\_m which the collective inertias should be calculated.

---

&MomentParam

---

MoreConstraints = Logical. If .true., signals that the namelist  
&MomentConstraint/ will follow.

ContinueAll = Logical. If .true., signal that the data of the constrained  
multipole moments should be taken from the input .wf file, not from STDIN.

A follow\_com = If .true., redefine the multipole moment operators at  
every iteration to follow the centre-of-mass of the nucleus  
around. Only active if parity is broken. Default: .false.

A MaxMoment = Maximum l of the multipole moments Q<sub>lm</sub> to be calculated. Default = 10

A MaxMomentMag = Maximum l of the magnetic moments to be calculated. Default = 3

A radd, acut, = Parameters of the cutoff function as pioneered by K. Rutz  
cutfac for multipole constraints. Defaults= (4 fm, 0.4fm, 10)

A cutofftype = Type of cutoff to use for the multipole constraints  
0: Density-dependent cutoff of the K. Rutz-type  
1: Spherical cutoff

---

```
&MomentConstraint    ATTENTION: * only active if MoreConstraints = .true.  
                      * this namelist can be repeated
```

|                   |   |                                                                                                                                                                                                          |
|-------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| l, m              | = | Integers, determining the multipole moment Q <sub>lm</sub> to constrain.                                                                                                                                 |
| Constraint        | = | Value to constrain <Q <sub>lm</sub> > to.                                                                                                                                                                |
| Iteration         | = | If -1 (=default), Tantalus will constrain the moment for the entire run. If other value, the constraint will only be used for this number of iterations. Useful for breaking self-consistent symmetries. |
| MoreConstraints   | = | If .true., signals that more &MomentConstraints namelists will follow. If .false., signals that this is the last occurrence of the &MomentConstraint namelist.                                           |
| Multfromfile      | = | Logical. If .true., start the calculation with the Lagrange multiplier of the constraint as read from file.                                                                                              |
| A iq1, iq2        | = | Real, alternative parameterisation of quadrupole moments. Cannot be combined with values for "Constraint". Needs to specify both a constraint on Q <sub>20</sub> and Q <sub>22</sub> to work.            |
| A Impart          | = | Logical. .false. if Real part, .true. if Imaginary part of the multipole moment. No versions of the code support Impart=.true.                                                                           |
| A Intensity       | = | Real. Intensity of this constraint in the penalty function. Default = 0.0. If left to default, the code will use a heuristic to estimate a proper value.                                                 |
| A Constrainttype  | = | Integer. How to handle this particular constraint:<br>0 : no constraint<br>1 : Augmented Lagrangian constraint (not recommended)<br>2 : Projection on feasible set constraint (Default)                  |
| A Scalefactor     | = | Real. Rescaling factor for the projection step in the case of constrainttype=2 constraints. Default=1.0                                                                                                  |
| A Intensityfactor | = | Real. Rescaling factor for the constraint intensity when obtained from heuristic. Default = 1.0.                                                                                                         |
| A Isoswitch       | = | Integer.<br>0 : Default, constraint on the total density (protons+neutrons)<br>1 : Constraint on the neutron density<br>2 : Constraint on the proton density                                             |

## &Cranking

```

omegaX/Y/Z      = Real, cranking frequency (hbar omega_x/y/z) in the x/y/z
                  direction in units of MeV. Non-zero values are not allowed
                  if time-reversal is not broken. Only omegaZ can take non-zero
                  values in all code versions discussed here.
crankX/Y/Z      = Real, target value for the expectation value of the angular
                  momentum <J_x/y/z> around the relevant Cartesian axis.
CrankTypeX/Y/Z  = Integer, determines the type of cranking constraint.
                  0 : Default, cranking at constant omega.
                  1 : Cranking constraint through projection on feasible space
                      Update of omega to obtain targetted J_x/y/z at convergence.
continuecrank   = Logical. If .true. use omegaX/Y/Z from the .wf file instead
                  of values read from STDIN.
A   IntensityX/Y/Z = Intensity of the cranking constraint for cranktype=1
                  calculations. Default = 0.0. If left to 0, the code will
                  use a heuristic to guess an appropriate value.
A   crank_smooth   = If .false. (=default), <J> is calculated for the readjustment
                  of the cranking constraints by summing the single-particle
                  expectation values of <J>, weighted with occupation factors.
                  If .true. <J> is instead calculated by integration of
                  current and spin densities.

```

### B. Output for the Brussels fitting codes

Output for the Brussels fitting codes is activated by entering a non-empty string for the BXLFIT parameter. The end-result is that Tantalus opens a file named \$BXLFITz\$Zn\$N.out and writes a single line to it at the end of the run.

```

      N, Z, Total energy, Enocorr, Erot+Evib,
&    b20, b22, b30, b32, b40                &
&    sqrt(<r^2_p>/Z), B(1:3), J2(1:3),       &
&    avgap_uv(n), avgap_uv(p), Epairn, Epairp &
&    DEhistory, iter, iomsg

```

The individual entries are

```

N,Z          = Average number of neutrons and protons          [MeV]
Total Energy = Total energy, including ALL terms.              [MeV]
Enocorr      = Total mean-field energy, no collective corrections. [MeV]
Erot + Evib  = Rotational and vibrational corrections.          [MeV]
blm          = Deformation values \beta_{ell m} for the total density
sqrt(<r^2_p>/Z) = Root-mean square charge radius                [fm]
B_x, B_y, B_z = Collective Belyaev moments of inertia around Cartesian axes [MeV^-1 hbar^2]
J2_x, J2_y, J2_z = Expectation value of <J^2> around Cartesian axes [hbar^2]
avgap_uv(n/p) = is the average gap for protons or neutrons as calculated for [MeV]
                the selected pairing approximation (HF/BCS/HFB) by either
                averaging the gaps with the anomalous density kappa (uv) for
                protons and neutrons.
Epairn,Epairp = Pairing energy of neutrons and protons          [MeV]
iter          = Number of SCF iterations performed by the code before exiting.
iomsg         = Output message by the code. Currently can be
                CONVERGED : the code exited because it judged convergence to be reached.
                MAXITER   : the code exited because it reached its maximum of iterations.
                CHECKPOINT: this wavefunction file was written as a checkpoint

```

### C. The structure of a parameterization (.param) file

The parameterization file \$name\_param.param contains all the values and special flags that characterize a parameterization. The structure of the .param file needs to match the executable, as not every form depends on the same parameters. This means that there are multiple type of .param files, although only one is available in the repository right now. It concerns:

- NLO EDF in the style of the Brussels group. Form as in G. Scamps et al., Eur. Phys. J. A 57, 333 (2021).
- ...

These are documented separately below. Note that, in all cases, the parameters that figure in the definition of the EDF are **REQUIRED** to be read from file. These parameters are indicated below with (R), all others can be omitted, in which case they will take (hopefully sensible) default values.

## 1. NLO EDF in the style of the Brussels group

```

&skf
name      = "SLy4"          (R) ! Name of the parameterization.
                                ! This needs to match name_param in the input data
func_file= 'NLO.func'      (R) ! Name of the functional file used for compilation.
t0        = -2488.913,      (R) ! t0 parameter of the Skyrme parameterization
t1        = 486.818,        (R)
t2        = -546.395,      (R)
t3        = 13777.0,        (R)
t4        = 0.0,            (R)
t5        = 0.0,            (R)
x0        = 0.834,          (R)
x1        = -0.344,         (R)
x2        = -1.0,           (R)
x3        = 1.354,          (R)
x4        = 0.0,            (R)
x5        = 0.0,            (R)
beta      = 0.0,            (R) ! Exponent of the density dependence (t4).
gamma     = 0.0,            (R) ! Exponent of the density dependence (t5).
sigma     = 0.1666667,      (R) ! Exponent of the density dependence (t3).
wso       = 123.0,          (R) ! Spin-orbit strength.
wsq       = 123.0,          (R) !
CT0       = 0.0,            (R) ! Coupling constant (isoscalar) of the J^2 terms
CT1       = 0.0,            (R) ! Coupling constant (isovector) of the J^2 terms
CSDS0     = 0.0,            (R) ! Coupling constant (isoscalar) of the s Delta s term
CSDS1     = 0.0,            (R) ! Coupling constant (isovector) of the s Delta s term
Vn        = -1250.00000,    (R) ! Pairing strength (neutrons) of the ULB interaction.
Vp        = -1250.00000,    (R) ! Pairing strength (protons) of the ULB interaction.
alpha     = 1.0,            (R) ! Alpha parameter in the ULB pairing interaction.
sigma_p   = 1.0,            (R) ! Power of the density dependence in the pairing interaction.
                                ! Alpha in EQ. (2) in S. Goriely et al, PRC88, 061302 (2013).
Estabp    = 0.0,            ! Cutoff factors for the stabilized pairing
Estabn    = 0.0,            ! functional of J. Erler et al., EPJA 37 (2008).
                                ! Set to zero to disable.
                                ! Typical value when active = 0.3 MeV

hbm(1)    = 20.73551910,    ! Value of hbar^2/2m
            20.73551910
force_switch = 0.0          ! INACTIVE
COM1Body   = 2              ! Treatment of the one-body COM correction.
                                ! (0) => None. (1) => Perturbative. (2) => Self-consistent
COM2Body   = 0              ! Treatment of the two-body COM correction.
                                ! (0) => None. (1) => Perturbative.
CoulTreatment = 1          ! Treatment of Coulomb interaction.
                                ! (0) => No Coulomb included.
                                ! (1) => Direct and exchange (Slater) contribution
                                ! (2) => Only direct contribution.

protonsize = 0.0, 0.0      ! Rms radii (r+, r-) of a double Gaussian model to be used for the proton.
                                ! If 0.0, no folding is used for either Gaussian.
neutronsize = 0.0, 0.0    ! Rms radii (r+, r-) of a double Gaussian model to be used for the neutron.
                                ! If 0.0, no folding is used for either Gaussian.
                                ! Note that input for protons and neutrons is NOT analogous.
                                ! The input for protons is the proton rms-radius, while for neutrons is
                                ! r_+^2 , r_-^2 in the model of Chandra & Sauer PRC13, 245
HOCOMform  = .false.       ! Correct the above rms radii for the c.m. correction in a HO basis?
nucleonsize_selfconsistent = .true./ ! Whether to use the folding self-consistently,
                                ! or only in the calculation of the energy.
rotcorr    = 0              ! Whether (1) or not (0) to include the rotational
                                ! correction and vibrational correction (!) in
                                ! calculation.
rotcorrb   = 0              ! Parameters b and c for the contribution of the
rotcorrc   = 0              ! rotational correction.
vibcorrb   = 0              ! Parameters b, d and l for the vibrational
vibcorrd   = 0              ! correction.
vibcorrl   = 0              !
eps        = 1d-20          ! Numerical parameter to safeguard the potentials
                                ! associated with density-dependent terms with
                                ! small exponents. Default value works for everything
                                ! without t4/t5 terms in my experience.

```

## V. APPLYING TRANSFORMATIONS

The code is capable of modifying a lot of aspects of information read from a .wf file. This section details a little bit of what is possible at the moment:

1. Modify the number of mesh points
2. Add single-particle wavefunctions
3. Break **a single** self-consistent symmetry

Important caveats are

- **Only one of these actions is allowed in any given run.** The code is hard enough to keep straight in that way, so I do not (and will not) support combinations in one go. If you want to do multiple things (say, break parity and add points along the z-axis), simply do them in successive runs.
- The code will only allow you to do any of these if `AllowTransform = .true.` in the `&IO/` namelist. Activating this trigger will remove a bunch of sanity checks on the input though, and it will be **very easy** to make mistakes.

Since it is very easy to do things wrongly, the code will always try to do its best to tell you what it actually did in the header, such as for example the following:

```

-----
| Transformation of the input
| On file:
|   nx, ny, nz   =   12   12   24
|   dx           =  0.8000000 (fm)
|   nwn, nwn     =    20    20
|   (n+,n-)      = (   20   0   0   0)
|   (p+,p-)      = (   20   0   0   0)
| This calculation:
|   nx, ny, nz   =   12   12   28
|   dx           =  0.8000000(fm)
|   nwn, nwn     =    20    20
|   (n+,n-)      = (   20   0   0   0)
|   (p+,p-)      = (   20   0   0   0)
|
-----

```

### A. Mesh modifications

Mesh modification is as simple as specifying the new mesh parameters in the input and setting `AllowTransform = .true.`. The code will transform the single-particle wavefunctions on the mesh by either cutting or adding points. Values of single-particle wavefunctions on any new point are initially set to zero. Note that the **user is responsible** for verifying that the box remains big enough if removing points; the code will do what you ask and not verify anything. Changing  $dx$  is perfectly possible, but does in my experience not gain anything in practice.

### B. Wavefunction modifications

Number of wavefunction modification is slightly more subtle: one should set `AllowTransform = .true.` and use the `extraspwfs` flag. This flag contains the extra wavefunctions as ordered in the symmetry-blocks of isospin/parity/signature.

```

extraspwfs = n++, n+-, n-+, n--, p++, p+-, p-+, p--
-----
|               neutrons               |               protons               |
|-----|-----|               |-----|-----|
| P = +1  | P = -1 |               | P = +1  | P = -1 |

```

